**Tutorial 09:** WRES1201 – Computer System Architecture

1. Give those characteristics distinguish RISC's organization.

- **One instruction per cycle:**
  Instructions are simple enough to be executed in a single machine cycle.
- **Register-to-register operations:**
  Almost all operations are performed within the CPU's registers. Memory is only accessed for explicit **LOAD** and **STORE** instructions.
- **Large number of registers:**
  To minimize memory access, RISC machines use many general-purpose registers or register windowing to store variables locally.
- **Simple and few addressing modes:**
  Complex address calculations are avoided to keep instruction decoding fast.
- **Fixed-length, simple instruction formats:**
  Every instruction is the same size (e.g., 32 bits), making fetching and decoding highly predictable and optimized for pipelining.

2. Briefly describe 2 basic approach used to minimize operation register-memory in RISC machine.

- **Software Approach (Compiler-based)**
  The compiler uses complex algorithms (like graph coloring) to carefully allocate which variables reside in the limited registers and which are sent to memory, maximizing register reuse over the variables' lifetimes.
- **Hardware Approach (Register Windows)**
  The CPU features a massive bank of registers, but the software only "sees" a small window of them at a time. When a function is called, the hardware simply shifts the "window" to a new set of registers, allowing local variables to be stored instantly without needing to push them to memory (the stack).

3. List and briefly define three types of computer system organization.

- **Symmetric Multiprocessors (SMP):** A system with multiple identical processors that share a single main memory and I/O facilities. They are controlled by a single OS and all processors have equal access and memory times.
- **Non-Uniform Memory Access (NUMA):** A shared-memory architecture where access time depends on the memory's location relative to a processor. A processor can access its own local memory faster than it can access shared memory located on a different processor's node.
- **Clusters:** A group of independent, whole computers (nodes) interconnected by a high-speed network. They work together to appear as a single unified computing resource.

4. What are the main characteristics of an SMP?

- There are two or more processors of comparable capability.
- These processors share the same main memory and I/O facilities.
- They are connected by a shared bus or other internal connection scheme.
- Memory access time is roughly identical for every processor (symmetric).
- All processors can perform the same functions (no strictly dedicated "boss" processor).
- The system is managed by a single, integrated operating system.

5. What is the difference between software and hardware cache coherent schemes?

- **Software Cache Coherence**
  Relies on the compiler and the operating system. The compiler analyzes the code to identify shared variables and inserts specific instructions to prevent caching of those shared items or to explicitly flush the caches. It requires less hardware but imposes a heavier computational burden on the software.
- **Hardware Cache Coherence**
  Relies on dynamic hardware mechanisms (like Snoopy or Directory protocols) that monitor memory traffic in real-time. It automatically updates or invalidates cache lines without any input from the compiler or OS. It requires complex hardware logic but is much faster and completely transparent to the programmer.

6. What is the meaning of each of the four states in the MESI protocol?

- **Modified (M):** The cache line contains data that has been modified. It is different from main memory, and this cache has the *only* valid copy.
- **Exclusive (E):** The cache line matches main memory and is present *only* in this specific cache.
- **Shared (S):** The cache line matches main memory and may also exist in other processors' caches.
- **Invalid (I):** The cache line does not contain valid data and must be fetched from memory if the CPU needs it.

7. What are some of the benefits of clustering?

- **Absolute scalability:** You can build a cluster with hundreds or thousands of nodes, vastly outperforming the largest single SMP machines.
- **Incremental scalability:** You can add new servers/nodes to the cluster one at a time as your computing needs grow.
- **High availability / Fault tolerance:** If one node in a cluster fails, the other independent nodes can take over its workload, preventing a total system crash.
- **Price/Performance:** Clustering relies on cheaper, off-the-shelf commodity computers rather than highly specialized and expensive mainframe hardware.

**Homework**

1. Let α be the percentage of program code that can be executed simultaneously by $n$ processors in a computer system. Assume that be remaining code must be executed sequentially by single processor. Each processor has an execution rate of $x$ MIPS.
   a. Derive an expression for the effective MIPS rate when using the system for exclusive execution of this program in term of α, $n$ and $x$.
   b. If n= 16 and x= 6 MIPS, determine the value of α that will yield a system performance of 54 MIPS.

2. *A program consists of five tasks, which have execution times of 2000, 4000, 6000, 8000 and 10,000 cycles. It is not possible to divide the execution of one task among multiple processors, but there are no communication or synchronization costs. If the task are distributed across the processor to achieve the shortest execution time, what is the speedup for executing the program on four processor?*

1. a) According to Amdahl's Law, speedup, $S = \frac{1}{(1-\alpha)+\frac{\alpha}{n}}$

Effective MIPS $= S \times x$

$$= \frac{x}{(1-\alpha)+\frac{\alpha}{n}}$$

b) $n=16$
$x = 6$ MIPS
Effective MIPS = 54 MIPS

$$54 = \frac{6}{(1-\alpha)+\frac{\alpha}{16}}$$

$$9 = \frac{1}{(1-\alpha)+\frac{\alpha}{16}}$$

$$(1-\alpha)+\frac{\alpha}{16} = \frac{1}{9}$$

$$-\frac{15}{16}\alpha = -\frac{8}{9}$$

$$\alpha = \frac{128}{135}$$

$$= 0.9481 \#$$

2. Single processor = 2000+4000+6000+8000+10000
= 30000 cycles

Multiprocessor : P1 : Task 5 (10000 cycles) ← parallel execution time
P2 : Task 4 (8000 cycles)
P3 : Task 3 (6000 cycles)
P4 : Task 2 + Task 1 (4000+2000 = 6000 cycles)

Speedup $= \frac{30000}{10000}$

$= 3.0 \#$